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Sorbent Intervention to Reduce Per- and Polyfluoroalkyl Substances (PFAS) in Maternal Reproductive Tissues and Transgenerational Translocation in Sprague-Dawley Rats

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Abstract

Per- and polyfluoroalkyl substances (PFAS) are persistent environmental contaminants, posing significant health risks, particularly during critical exposure windows such as gestation and lactation. In this study, following prenatal and postnatal exposure to 0.95 mg PFAS/kg-bw/day in F₀ Sprague-Dawley rats, their toxicity to maternal reproductive tissues and translocation to offspring were measured. On postnatal day (PND) 21, PFAS (especially PFOS) were detected in F₀ ovaries and testes with histopathological ovarian damage. PFAS time-dependently accumulated in maternal milk and transferred approximately 1% to F₁ serum and 3.7-6.0% to F₁ livers, except for PFBS. PFAS exposure reduced offspring birth weight and growth by 20.8% ($p \leq 0.01$), and induced systemic inflammation and hepatotoxicity, as shown by the imbalanced liver function parameters, antioxidants, and pro-inflammatory cytokines. These findings suggest that long-term health implications may result from early-life PFAS exposure.

To mitigate PFAS toxicity, generally recognized as safe (GRAS) sorbents were orally administered during PFAS exposure. Activated carbon (AC) and acid-processed montmorillonite (APM) treatments most significantly reduced total PFAS in maternal ovaries by 47-49%, testes by 40-43%, and milk by 40-74% ($p \leq 0.01$), and restored ovarian histoarchitecture. Although sorbents only slightly reduced PFAS concentrations in the offspring liver and serum by 5%, they prevented

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Johnson O. Oladele: Writing – original draft, Investigation, Formal analysis. **Kelly J. Rivenbark:** Writing – review & editing, Investigation. **Damilare A. Adekomi:** Writing – original draft, Investigation. **Timothy D. Phillips:** Writing – review & editing, Validation, Funding acquisition. **Meichen Wang:** Writing – original draft, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

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PFAS-induced oxidative stress, inflammation, and liver damage. These findings emphasize the urgent need for targeted short-term interventions to reduce PFAS exposure and toxicity, especially during vulnerable periods. Enterosorbent treatments offer a promising preventive approach by reducing the systemic bioavailability of PFAS and interrupting toxic effects in two generations.

Keywords

Reproductive toxicity; Pregnancy; Clay; Carbon; Adsorption; Offspring; Exposure

1. Introduction

Per- and polyfluoroalkyl substances (PFAS), especially the studied perfluoroalkyl acids and perfluoroalkylether acid, are synthetic chemicals widely used in industrial and consumer applications for their unique water- and lipid-repellent properties. Their chemical stability leads to environmental persistence and bioaccumulation in organisms, raising health concerns and urging regulatory attention to cumulative, mixture-based risks (Huang et al., 2019, Wang et al., 2019). For example, while legacy PFAS such as perfluorooctanesulfonic acid (PFOS) and perfluorooctanoic acid (PFOA) account for the most exposure to newborns, emerging PFAS such as perfluorobutane sulfonic acid (PFBS) and hexafluoropropylene oxide dimer acid (HFPO-DA, GenX) are being detected more frequently in human populations (Zheng et al., 2022). Therefore, more studies are necessary to explore the potential health impacts and mitigation of both legacy and emerging PFAS.

Human exposure to PFAS occurs through multiple pathways, with oral ingestion being a primary route (Zheng et al., 2022, Roth et al., 2020). Pregnant and lactating individuals and their infants are especially vulnerable because prenatal PFAS exposure has been associated with adverse outcomes such as gestational hypertension, low birth weight, and impaired liver function in offspring (Zheng et al., 2022, Jung et al., 2024, Blake and Fenton, 2020). These effects can be a result of PFAS accumulation in maternal reproductive tissues, including the ovaries and mammary glands, and transfer to offspring through the placenta and breast milk (Blake and Fenton, 2020).

Despite the well-documented toxicity of PFAS, there is a notable lack of preventive strategies to reduce exposure and mitigate adverse health effects in animals and humans (Shao et al., 2021, National Academies of Sciences Engineering and Medicine 2022, TCIM 2024). Although environmental remediation is important, and activated carbon (AC) has been widely studied to adsorb PFAS in wastewater treatment, its efficacy in reducing dietary exposure and *in vivo* body burden remains poorly characterized. Current interventions for *in vivo* PFAS exposures include bile acid sequestrants and dietary supplements that can bind PFAS and promote fecal excretion (Bonanni and Newman, 2024, Stein and Heimann, 1975, Mayo Clinic n.d., Sunderland et al., 2019). However, their clinical use is limited by side effects, nutrient malabsorption, and interference with medications (Mayo Clinic n.d., Dzierlenga et al., 2021).

As a complementary preventive approach, orally administered sorbents that sequester contaminants in the gut (enterosorbents) offer promise for reducing systemic uptake

during exposure windows. Acid-processed montmorillonite (APM) was developed to simulate carbon with high porosity and surface area and was patented for its enhanced binding efficacy and safety (Phillips and Wang, 2021). Furthermore, we have modified montmorillonite clays, recognized as safe for human consumption (Phillips et al., 2019), with natural nutrients such as L-carnitine and choline. These functionalized clays demonstrated high efficacy in removing PFAS mixtures from water and soil and reducing plant uptake (Wang et al., 2024, Hearon et al., 2022, Wang et al., 2021). Our previous human clinical trials against aflatoxin have supported the use of sorbents as a preventive mitigation strategy during exposure (Phillips et al., 2019, Afriyie-Gyawu et al., 2008, Phillips and Carpenter, n.d., Wang et al., 2005), where high-affinity sorbents at 0.25% and 0.125% (w/w) per meal (3 meals/day) effectively sequestered toxic chemicals in the gastrointestinal tract and thus prevented systemic exposure and toxicity.

These findings support the design of this study, which was to orally administer sorbents during susceptible windows of exposure, such as pregnancy and lactation, to reduce PFAS oral exposure and prevent reproductive toxicity and transgenerational transfer. Rats were used for their consistent toxicokinetic data for PFAS (Chou and Lin, 2019) and large sample volumes for comprehensive analysis. F₀ adult male and pregnant female rats were exposed to a PFAS mixture and treated with different sorbents during gestation and lactation. F₁ offspring were only indirectly exposed. The study aims to 1) quantify PFAS concentrations in maternal reproductive tissues, milk, and pups, 2) assess PFAS reproductive toxicity and outcomes in offspring, and 3) compare sorbent efficacy in reducing maternal body burden and transgenerational transfer.

2. Materials and Methods

2.1. Chemicals and reagents

The target PFAS analytes, including perfluorooctanoic acid (PFOA, CAS 335-67-1), perfluorooctanesulfonic acid (PFOS, CAS 1763-23-1, >95%), hexafluoropropylene oxide dimer acid (HFPO-DA, GenX, CAS 13252-13-6), and perfluorobutane sulfonic acid (PFBS, CAS 375-73-5, >98%) were purchased from Sigma-Aldrich (St. Louis, MO). A PFAS mixture containing equal concentrations of each PFAS was used in all studies. Three isotopically labeled internal standards (IS): ¹³C₄-PFOA, ¹³C₃-HFPO-DA, and ¹³C₄-PFOS (purity >99%, Wellington Laboratories, Guelph, ON) were employed for quantification of PFOA, GenX, PFOS and PFBS, respectively. The PFAS mixture was dissolved in deionized saline water and prepared fresh on the day of gavage.

Calcium montmorillonite (CM) clay [(Na,Ca)_{0.3}(Al, Mg)₂Si₄O₁₀(OH)₂·nH₂O, BASF, Germany] from our previous human clinical trials was included as the base material for clay modifications. Organoclays resulting from the amendments of L-carnitine and choline at 100 % cation exchange capacity of 89.2 cmol/kg (Phillips et al., 2019, Wang et al., 2017, Grant and Phillips, 1998) were previously developed (Wang et al., 2021, Wang et al., 2019). APM clay was synthesized by treating CM at 6% (w/v) with sulfuric acid at 12 normality (Wang et al., 2019, Wang and Phillips, 2019), yielding a mixture of delaminated clay layers with amorphous and cross-linked silica (Wang et al., 2019, Wang and Phillips, 2020). A virgin, medical-grade AC derived from coconut shell (purity >99%, General

Carbon Corporation, Paterson, NJ) was used in this study for its high surface area and carbon content (Hearon et al., 2022, Wang et al., 2021). These sorbents have been previously synthesized or purchased, characterized for surface area, pH_{pzc} (point of zero charge), density, moisture, water expansibility, hydrophobicity, particle size, and analyzed by SEM (Scanning Electron Microscopy), XRD (X-Ray Diffraction), and FT-IR (Fourier Transform Infrared Spectroscopy) (Wang et al., 2024, Grant and Phillips, 1998, Wang et al., 2021, Wang et al., 2021).

2.2. Treatment of F₀ rats

All animal studies were conducted with approval by the Institutional Animal Care and Use Committee (IACUC) of Texas A&M University (IACUC 2021-0327). The overview of the study design is illustrated in Fig. 1.

Male and female Sprague-Dawley rats (8–10 weeks old, 200–300 g) were obtained from Charles River Laboratories (Raleigh, NC). Animals were acclimated and socialized for 7 days in an AAALAC-accredited facility (College Station, TX) under controlled environmental conditions at 22.2–23.9°C, 50% relative humidity, and a 12-hour light-dark cycle. Following acclimation, rats were randomly assigned to treatment groups using computerized random-sequence generation. On gestational day (GD) 1, the standard deviations of the group mean weights were 1.8% for females and 6.2% for males. The rats were housed individually in ventilated polycarbonate cages and provided a standard pellet diet (Teklad Rodent Diet 8604, Harlan, WI) and RO water *ad libitum*. Animal cages and treatments were coded throughout dosing, sample collection, and data analysis to ensure that operators remained blinded to treatment groups.

For mating, one male was cohabited with one or two females until confirmation of mating, which was evidenced by the presence of spermatozoa in vaginal smears and/or a copulatory plug, marking as GD 0. Females that failed to conceive were excluded before the treatment started, resulting in at least 5 pregnant females that successfully delivered offspring, along with 4 males being included per group by the end of the study (n = 4-7/sex/group).

Adult F₀ rats received an oral PFAS mixture consisting of PFOA, PFOS, PFBS, and GenX, each administered at a dose of 4.75 mg/kg body weight (bw)/gavage. This dose, translates to 0.95 mg/kg-bw/day in rats or 0.16 mg/kg-bw/day in humans (Nair and Jacob, 2016), was selected as the lowest level to detect toxicity of each PFAS and to ensure maternal-fetal transfer (Marques et al., 2021). The selected dose was based on the no observed adverse effect level (NOAEL) of GenX (1 mg/kg/day) (Caverly Rae et al., 2015). We intentionally used equal concentrations of PFOA, PFOS, PFBS, and GenX in the defined mixture to compare differences of each PFAS in bioavailability and half-lives and to minimize dose heterogeneity as a confounder (Garmo et al., 2024, Schlezinger et al., 2025, Maxwell et al., 2024, Roth et al., 2024, Umeh et al., 2024, Carberry et al., 2024, Haange et al., 2024, He et al., 2025, Sun et al., 2025). PFAS solutions were prepared in saline water (pH 7-8) according to body weight on the day of administration. Sorbents, including CM-carnitine, CM-choline, AC, APM, and CM, were co-administered at 2% (w/v) during gavage. This concentration aligns with the maximum level (2.5% w/w) of clays permitted by the Association of American Feed Control Officials (AAFCO) (McKenzie et al., 1997).

for use as an anti-caking agent in non-medicated livestock feeds. PFAS, with and without sorbent inclusion, was administered by oral gavage at a volume of 2.5 mL/kg on GD 9, GD 14, GD 19, postnatal day (PND) 9, PND 14, and PND 19. Dosing began on GD 9 to confirm pregnancy and exclude non-pregnant females before the experimental exposure. The 5-day dosing interval (Liu et al., 2023, dos S. Barros et al., 2024, Queiroz et al., 2025) was used to reduce our interaction with rats during gestation and lactation, and minimize procedure-related stress such as increased blood pressure, heart rate, and plasma corticosterone levels (Walker et al., 2012, Hoggatt et al., 2010, Brown et al., 2000, McDonnell-Dowling et al., 2017, Eichenbaum et al., 2011). Vehicle control rats received an equivalent volume (2.5 mL/kg) of saline water.

2.3. Measurement of F₁ pups

The dams were checked daily prior to the end of the light cycle to identify the delivery date of the litter (PND 0). The weight and number of pups were measured on PND 1 and every 5 days thereafter. The number of live or dead fetuses and resorptions was recorded. Throughout the lactation period, sex of the pups was determined, and pups were examined for external abnormalities. All F₁ pups were euthanized on PND 21, and blood and liver were individually collected and stored. PND 21 was selected as the time point for F₁ pups based on our preliminary data, where PFAS levels at PND 1 and 10 were similar to the vehicle.

2.4. Sample collection

To collect milk from dams, pups were separated into a box located in the same cage for 4 hours to allow milk accumulation. Dams were then anesthetized under isoflurane gas flow connected through a nose cone and intraperitoneally injected (i.p.) with oxytocin (2 IU/kg) (Shore et al., 2024). Milk collection was performed by gentle manual massage of the mammary glands to pump milk and aspirated by pipetting into a microcentrifuge tube (Muranishi et al., 2016). Milk samples were collected from all dams on PND 1, PND 10, and PND 21.

F₀ parents were euthanized two days after the last administration (PND 19). On PND 21, euthanasia of adult rats was performed using 70% CO₂/30% O₂, following the standard AVMA Guidelines. Ovaries and testes were harvested from F₀ rats, weighed, measured, and stored at -80°C for subsequent chemical and histopathological analysis. F₁ pups were also euthanized on PND 21, and the liver was individually collected and kept at -80°C. Up to 3 mL of blood was immediately collected via cardiac puncture from each pup. A 23-gauge needle connected to a 10 mL syringe was inserted posterior to the xiphoid cartilage and slightly to the left. Whole blood was allowed to clot at room temperature for 3 hours, followed by an additional 3 hours at 4°C. Serum was isolated via centrifugation at 2,000 rpm for 10 min at 4°C, repeated twice, and stored at -80°C for further analysis.

2.5. PFAS extraction and quantification by LC-MS/MS

Samples analyzed for PFOA, PFOS, GenX, and PFBS concentrations include maternal milk on PND 1, 10 and 21; maternal ovaries and testis on PND 21; and offspring serum and liver on PND 21.

Thawed serum samples (100 μ L/sample) were spiked with 0.25 μ M 13 C-PFAS IS and denatured by adding 300 μ L of cold (-20°C) acetonitrile containing 0.1% formic acid in 1.5 mL centrifuge tubes. Samples were shaken at 300 rpm for 20 min and centrifuged at 4,000 rpm for 10 min to precipitate the serum proteins. A 350 μ L aliquot of the resulting supernatant was diluted with 1.5 mL of 1% formic acid. Sample cleanup was performed using a Waters Oasis Weak Anion Exchange (WAX) solidphase extraction cartridge (SPE, 150 mg/6 mL, Bedford, MA), which was preconditioned sequentially with 0.3% ammonia in methanol, methanol, and 1% (v/v) formic acid. Each sample was loaded onto the SPE cartridge twice to ensure maximal retention. Tubes and SPE cartridges were washed with 1% formic acid and 30% acetonitrile, respectively. PFAS analytes were eluted using methanol and 0.3% ammonia in methanol, and the eluates were evaporated to near dryness under a gentle stream of nitrogen (Organtini et al., 2022). Dried residues were reconstituted in 40% methanol and 60% acetonitrile and filtered through 0.2 μ m nylon syringe filters (Sartorius) for analysis.

Thawed tissue samples of ovary, testis, and liver (0.5 g/sample) were homogenized in water at a 3:1 (v/w) ratio using a 150 Homogenizer and 7 mm Generator Probe (Fisher Scientific, Waltham, MA). Homogenized tissue samples were spiked with 0.25 μ M 13 C-PFAS IS and extracted with 2 mL of methanol containing 0.02 M sodium hydroxide. Samples were shaken at 500 rpm for 1 hour and sonicated for 30 min. After centrifugation at 4,000 rpm for 10 min, the supernatant was collected and diluted with 3 mL of water (pH 2, adjusted with formic acid). Sample clean-up was performed using Waters OASIS WAX SPE, which was preconditioned with 4 mL each of 0.3% (v/v) ammonium hydroxide in methanol, methanol, and 1% (v/v) formic acid. Liver extracts were loaded onto the SPE cartridge twice to maximize analyte recovery. Tubes were rinsed with 4 mL of 25mM ammonium acetate (pH 4) and methanol. SPE was eluted with 5mL of 2% (v/v) ammonium hydroxide in methanol, and the eluates were dried at 50-60°C until one drop remained. Samples were then reconstituted in 0.5 mL of 1:1 (v/v) 2 mM ammonium acetate:methanol and filtered through 0.2 μ m nylon syringe filters for instrumental analysis.

Extraction of milk was performed following QuEChERS extraction and dSPE clean-up protocols (Vaclavik et al., 2022). Briefly, milk samples (0.1 mL/sample) were spiked with 0.25 μ M PFAS IS and diluted with 0.1 mL each of water and acetonitrile containing 1.5% formic acid and mixed for 1 min. QuEChERS salt mixture (UCT, Hayward, CA) was added at 0.2 g and immediately shaken to ensure that crystalline agglomerates were broken up sufficiently and vortexed for 5 min. Samples were centrifuged at 3,000 g for 5 min, and supernatants were added to dSPE (UCT, Hayward, CA) for 2 min and then centrifuged at 3,000 g for 5 min. Final supernatants were evaporated to near dryness under a gentle stream of nitrogen and reconstituted in methanol and filtered through nylon 0.2 μ m filters into polypropylene autosampler vials for LC-MS/MS analysis.

Eluates containing the 4 target PFAS were analyzed by liquid chromatography tandem mass spectrometry (LC-MS/MS) with an Acquity BEH C18 column (2.1 $\times$ 50 mm, 1.7 μ m) maintained at 40°C (Heaton et al., 2022, Wang et al., 2021, Wang and Phillips, 2022). Quantitation was based on recoveries of IS. A binary gradient elution was performed using 20 mM ammonium acetate (eluent A) and acetonitrile (eluent B) at a flow rate of 0.6

mL/min. The gradient program was as follows: 10% eluent B (initial), 10%–55% (0 to 1 min), 55%–99% (1 to 9 min), 99% (9 to 10 min), and 99%–10% (10 to 13 min) for column re-equilibration. The injection volume was 10 μ L for each analysis. Detection was performed using electrospray ionization (ESI) in negative ion mode with a spray voltage of –4.5 kV and a source temperature of 450°C. The mass spectrometer operated in multiple reaction monitoring (MRM) mode for specific PFAS transitions. Data acquisition and instrument control were managed using Empower software (Waters Corporation). Quality assurance/quality control procedures and PFAS-specific detection parameters are provided in the Supplementary Material.

2.6. Histopathology

F₀ ovaries collected on PND 21 were fixed in 10% neutral buffered formalin for 72 hours, followed by paraffin embedding. Tissue sections were cut at a thickness of 5 μ m using a microtome and stained with hematoxylin and eosin (H&E). Histological images were captured using a light microscope (Olympus, Tokyo, Japan). All sections were examined and validated by a board-certified veterinary pathologist for the surveillance of any treatment-related histomorphological changes.

2.7. Markers

The glutathione (GSH) antioxidant activities in the liver and serum of pups were quantified by ELISA assay kits (Cayman, Ann Arbor, MI). Serum inflammatory cytokines, including alanine aminotransferase (ALT), aspartate aminotransferase (AST), transforming growth factor beta (TGF- β), interleukin-6 (IL-6), tumor necrosis factor-alpha (TNF- α), and C-reactive protein (CRP) were measured using commercially available assay kits (Cayman, Ann Arbor, MI).

2.8. Statistical analysis

Data are presented as means and standard deviations. Statistical significance was determined by one-way analysis of variance (ANOVA) followed by Tukey's post-hoc test using OriginPro software. Individual litter was used as the statistical unit. Statistically significant differences were determined at $p \leq 0.05$.

3. Results and Discussion

This study investigated PFAS toxicity in maternal reproductive organs, transfer to offspring, as well as the efficacy of sorbent-based mitigation strategies. The study included 28 F₀ males, 38 F₀ dams, and 38 corresponding F₁ litters. F₀ adults were exposed to one of seven treatment conditions throughout gestation and lactation, including vehicle control, PFAS control (4.75 mg/kg bw/gavage), PFAS + CM-carnitine, PFAS + CM-choline, PFAS + AC, PFAS + APM, and PFAS + CM ($n = 4\text{--}7/\text{sex}/\text{group}$). The milk of F₀ dams was collected on PND 1, 10, and 21 for PFAS quantification. The ovary and testis of F₀ parents, as well as the liver and serum of F₁ pups, were collected on PND 21 for PFAS concentration and toxicity analyses (Fig. 1).

3.1. Clinical signs of F₀ rats

PFAS exposure induced sex-specific changes in reproductive organ weights (Fig. S1, Table 1). In F₀ females, relative ovary-to-body weight decreased by 30.3% (from 0.49% to 0.34%, $p \leq 0.01$), aligning with prior studies showing PFAS-triggered follicular atresia and granulosa cell apoptosis via oxidative stress and estrogen receptor antagonism (González-Alvarez et al., 2024). Conversely, testis-to-body weight increased by 20.9% (from 0.95% to 1.15%, $p \leq 0.05$), a paradoxical effect that has been previously observed in animal studies (Caverly Rae et al., 2015). Specifically, decreased testes have been observed in development and high-dose studies, where PFAS was suggested to impair testis growth and cause inflammation and apoptosis (ATSDR, 2021); while other studies showed increased testes weight, possibly by activating steroidogenic genes and steroid hormones (Han et al., 2022). The changes in the relative weight are suggestive of PFAS impact on reproductive organs.

Sorbents administered at 2% showed trends to alleviate these changes in both absolute organ weights and relative organ weight to body weight. APM most effectively reversed organ weight changes by significantly restoring ovarian weight to near-control levels ($p \leq 0.05$) (Table 1). The binding mechanisms may be due to APM's porous structure containing mesopores and micropores, its high surface area (Wang et al., 2021), and potential electrostatic interactions between protons on intact APM interlayer surfaces and anionic PFAS (Hearon et al., 2022, Wang et al., 2019).

3.2. PFAS in the ovary and testis

All 4 PFAS were detected in the reproductive organs in F₀ female and male rats on PND 21, as a result of accumulation of PFAS exposure during gestation and lactation (Fig. 2). PFAS levels in the ovary were 1.2 to 2.1 times higher than in the testis ($p = 0.36$). Long-chain PFOS dominated tissue accumulation and was detected at approximately 5 times higher concentration than other PFAS ($p \leq 0.05$). This finding aligns with the longer serum half-life of PFOS at 20-30 days (Conley et al., 2022, Kim et al., 2016, Chang et al., 2012) and its stronger protein-binding affinity. Low levels of PFAS, indicative of background environmental contamination, were detected in vehicle control groups. However, their concentrations were significantly lower than those in PFAS-exposure groups.

Among the various sorbent treatments, AC and APM were the most effective in reducing PFAS bioavailability in reproductive organs. On average, these sorbents reduced GenX by 59% and 16%, PFBS by 41% and 47%, PFOA by 30% and 30%, and PFOS by 36% and 44% in the ovary and testis, respectively. The mechanism of sorbent protection involves sequestering PFAS in the gastrointestinal tract, thus limiting their systemic absorption and circulation in the body. AC and APM possess similar porous structures with mesopores and micropores, which enhance their capacity to adsorb both short- and long-chain PFAS.

3.3. Histopathology of the ovary

The ovaries were evaluated for histopathology because of the higher PFAS accumulation and toxicity in the ovaries than testes. H&E stained ovarian sections from the vehicle control group showed intact histological architecture, with no observable abnormalities in the germinal epithelium, tunica albuginea, or cortical stroma (Fig. 3A). Several ovarian follicles

(arrows in Fig. 3A) at various stages of maturation and several corpora lutea were evident, and the ovarian medulla showed normal vascularity. The oocytes were intact with normal histological profiles of the granulosa and theca cell layers. In contrast, PFAS exposure showed histopathological alterations indicative of ovarian dysfunction, with the presence of numerous large subcapsular cystic follicles along with dilated, congested blood capillaries (Fig. 3B), which suggested polycystic ovary syndrome (PCOS). Additionally, there are fewer corpora lutea observed, and most follicles remained in early developmental stages. The granulosa cell layers were sparse and loosely organized, accompanied by thickening of the outer theca layer. These observations align with previously reported PCOS following PFAS exposures, and this study further indicates the ability of PFAS to disrupt the estrous cycle, induce hormonal imbalance, and cause infertility.

Interventions with adsorbents mitigated PFAS-induced damage to varying degrees. Treatment with AC in Fig. 3C demonstrated a restoration of follicular development, showing corpora lutea integrated within the cortical stroma. Likewise, the ovarian sections showed enhanced histoarchitecture of the granulosa and theca cell layers. Treatment with APM demonstrated comparable efficacy, with mature Graafian follicle, ovarian follicle, and corpus luteum (c in Fig. 3D). Furthermore, an oocyte (o in Fig. 3D) with a distinct zona pellucida (arrow in Fig. 3D), several layers of granulosa cells, and theca cells were observed. The ovarian stroma is merging with the spindle-shaped cells of the theca externa. The polyhedral cells border the follicular cavity and surround the oocyte. Numerous granulosa lutein cells were found in the corpus luteum. Fine collagen fibers, spindle-shaped cells, fibroblast-like cells, and ground material were all present in the ovarian stroma. Treatment with CM exhibited ovarian follicles with a predominantly normal histoarchitectural profile (Fig. 3E). Specifically, the follicular structure was intact with densely packed granulosa cells forming the follicular wall. The epithelial lining surrounding the follicles is well-defined and continuous without disruption, cellular degeneration, or inflammation (Fig. 3E). Enhanced effects were observed with CM functionalized with carnitine and choline, where the histoarchitectural profile of the ovary appeared normal with the surface epithelium layer surrounding the ovary (red arrows in Fig. 3F & G). The ovarian stroma contained numerous primordial follicles (black arrows). The oocytes and zona pellucida in the primary follicles were surrounded by a single layer of cuboidal granulosa cells and an outer layer of flattened cells (Hammarstrand et al., 2021, Wang et al., 2019, Zhang et al., 2024) (yellow arrows) and antral follicle (white arrow), indicating resumed folliculogenesis.

3.4. PFAS in milk

Analysis of PFAS concentrations in F₀ dam milk revealed significant time-dependent and compound-specific accumulation patterns. Fig. 4A-D shows the average PFAS levels calculated from data on PND 1, 10, and 21, accounting for the higher data deviation. Similar to results in reproductive organs, PFOS dominated milk contamination, peaking at 38.7 ng/mL on PND 21 and averaging at 22.7 ng/mL throughout lactation, followed by PFOA, GenX, and PFBS ($p \leq 0.01$). These trends correlate with PFAS physicochemical properties, where long-chain PFOS (C8) and PFOA (C8) exhibited higher lipophilicity and protein-binding affinity, enhancing their sequestration into lipid-rich milk matrices. Short-chain GenX (C6) and PFBS (C4) showed lower but persistent transfer, likely due to

their higher water solubility and faster renal excretion. For most PFAS, their levels in the milk were about 1 order of magnitude lower than those in dam serum (Wang et al., 2025), aligning with literature (Zheng et al., 2022).

All 4 PFAS (GenX, PFBS, PFOA, and PFOS) were detected in milk across PND 1, 10, and 21, with levels escalating progressively during lactation (Fig. 4E). Especially, PFOS concentration increased 5-fold from 7.6 ng/mL on PND 1 to 38.7 ng/mL on PND 21 ($p \leq 0.01$), compared to approximately 2-fold of increase for other PFAS. This time-dependent escalation was a result of repeated maternal exposure during gestation and lactation and the progressive partitioning of PFAS into mammary tissue lipids, which expand during lactation.

All of the sorbent treatments significantly reduced total PFAS in milk, with the reduction rates ranging from 36-74% ($p \leq 0.05$). Similar to previous results, AC was the most effective sorbent that reduced GenX by 76%, PFBS by 77%, PFOA by 83%, and PFOS by 59% ($p \leq 0.01$). The levels of PFAS were reduced to levels similar to vehicle controls, suggesting high efficacy of mitigation.

3.5. Birth outcome and growth

All neonates in all groups were born alive and appeared active. The majority successfully survived through the lactation period, and the survival rates at weaning were comparable among groups with 96.5% for vehicle, 97.2% for PFAS, 96.7% for CM-carnitine, 98.8% for CM-choline, 95% for CM, and 100% for APM and AC. There was also no obvious difference in maternal nesting behavior, sex distribution of litters, or malfunction observed among different treatments. It is possibly because the PFAS dosage in this study (0.95 mg/kg bw/day) was lower than administered doses in the literature that caused mortality in rodents, such as 20 mg/kg of PFOA (Lau et al., 2006) or PFOS (Stub et al., 2026). However, maternal exposure to PFAS significantly reduced pup birth weight to 6.1 ± 0.4 g compared to vehicle controls (6.9 ± 0.4 g, $p \leq 0.05$) (Fig. 5). PFAS has also significantly reduced growth that by the end of PND 19, the average pup weight in PFAS groups (24.4 g) was 20.8% lower than vehicle controls (30.8 ± 2.7 g, $p \leq 0.01$) (Fig. 5). This developmental toxicity of PFAS in offspring correlates with the accumulation of PFAS, particularly long-chain PFOS, in breast milk in Fig. 4. Low birth weight and slow growth in offspring are the most consistently reported adverse pregnancy outcome associated with gestational exposure to PFAS in human epidemiologic and animal studies (Jung et al., 2024, Walker et al., 2012, Hoggatt et al., 2010, Brown et al., 2000).

However, in a separate study where offspring were kept for 66 days till sexually mature, the weight difference between PFAS-treated (265 g) and vehicle groups (270 g) diminished, which demonstrates a catch-up and recovery effect in adulthood when exposure ceased post-weaning. It was previously reported that prenatal or early-life exposure to PFAS (e. g., PFOS) is associated with reduced birth weight but increased susceptibility to obesity in adulthood (Lv et al., 2013). This "catch-up growth" may involve altered adipogenesis and insulin resistance (Zatterale et al., 2020, Jackson et al., 2017).

Importantly, all sorbent treatments significantly improved pup growth compared to PFAS exposure ($p \leq 0.01$) (Fig. 5). Interestingly, pup growth in AC and APM even exceeded vehicle controls on PND 19 (37.7 g and 39.5 g vs 30.8 g). Additionally, treatment with AC increased pup birthweight to 7.1 ± 0.5 g, similar to that of vehicle control (6.9 ± 0.4 g) at birth. This result suggests that the prevention of sorbents by blocking PFAS uptake in the maternal gastrointestinal tract and reducing PFAS circulation and accumulation in reproductive organs (Fig. 2) and milk (Fig. 4) has a long-lasting effect even in the offspring who were indirectly exposed through breast milk.

3.6. PFAS translocation to offspring serum and liver

Since PFAS was detected in maternal milk, to investigate the translocation of PFAS to F_1 offspring, their serum and liver were collected on PND 21 for PFAS concentrations. PFBS was not detected in the offspring, possibly due to its short half-life and low concentrations in the dam's milk (Fig. 4). PFOS was detected at approximately 10 times higher than other PFAS (Fig. 6), possibly because of higher PFOS levels in maternal milk (Fig. 4). All PFAS accumulated more in the liver than in serum, with liver-to-serum ratios of PFOA, PFOS, and GenX at 1.7, 5.2, and 3.3, respectively. This preferential accumulation in the liver is likely due to the higher affinity of PFAS for protein, such as fatty acid-binding proteins and albumin, in hepatic tissues. The longer half-life of PFAS in the liver than in serum is another possible reason for the higher concentrations in the liver. The transfer rate of PFAS from maternal milk to offspring serum and liver on PND 21 was calculated and found to be highly comparable across different PFAS. Specifically, about 1% PFAS in maternal milk was transferred to F_1 serum, while $4.5 \pm 0.29\%$ GenX, $3.7 \pm 0.21\%$ PFOA, and $6.0 \pm 1.28\%$ PFOS were transferred to F_1 liver on PND 21.

Treatments with sorbents, which were administered to F_0 rats, slightly reduced total PFAS transfer to offspring by approximately 5%. The reduction rates were much lower in F_1 compared to those in F_0 , because sorbents were only indirectly administered to offspring. Direct administration of sorbents to F_1 pups post-weaning may be needed for higher mitigation efficacy, depending on the exposure level.

3.7. PFAS impact on immunity markers

F_1 offspring on PND 21 were analyzed for key inflammatory cytokines and antioxidant levels in the liver and serum to assess the systemic effects of PFAS exposure. The results revealed significant alterations in markers of oxidative stress, liver function, and immune response, suggesting immunity as one of the mechanisms of PFAS-induced toxicity.

PFAS exposure resulted in a reduction of glutathione (GSH) levels in both serum and liver (Fig. 7A). GSH is a critical intracellular antioxidant responsible for neutralizing reactive oxygen species (ROS) and protecting cells from oxidative damage. The depletion of GSH indicates elevated oxidative stress and a compromised antioxidant defense system. Given that the liver is a major site of PFAS accumulation and metabolism, it is particularly vulnerable to such oxidative damage, which may contribute to the observed change in GSH in the liver. Additionally, significant elevations in alanine aminotransferase (ALT) and aspartate aminotransferase (AST) were observed in PFAS-exposed offspring (Fig. 7B and

7C). These enzymes are commonly used as biomarkers of liver damage, as they are released into the bloodstream upon hepatocyte injury or necrosis. The increased levels of ALT and AST indicate that PFAS exposure induced hepatotoxicity. These findings are consistent with previous studies linking PFAS exposure to liver damage, including steatosis, inflammation, and fibrosis (Wang et al., 2025, Costello et al., 2022).

Furthermore, PFAS induced inflammatory cytokines, including transforming growth factor-beta (TGF- β) (Fig. 7C) in the serum, indicating fibrotic changes in the liver. Additionally, the increase in pro-inflammatory markers such as tumor necrosis factor-alpha (TNF- α) and interleukin-6 (IL-6) indicates a systemic inflammatory response. Interestingly, C-reactive protein (CRP) levels were downregulated following PFAS exposure, suggesting a potential immunosuppressive effect. This reduction may be attributed to PFAS-induced liver damage, which could impair the hepatic synthesis of CRP. Collectively, these findings indicate that PFAS exposure is associated with systemic inflammation and hepatic dysfunction.

Most sorbent treatments ameliorated these changes to different extents (Fig. 7). All sorbents significantly reduced PFAS impact on GSH and CRP in serum ($p \leq 0.05$). Treatment with CM-choline most effectively reversed changes of GSH in the liver by 33.5%, and increased AST by 17.6% and ALT by 48.9%. Whereas the parent CM was the most effective sorbent in reducing TGF- β by 15.7%, TNF- α by 23.8%, and IL-6 by 20.3%. These differences between sorbents suggest that a comprehensive analysis is warranted to understand the complex nature of PFAS toxicity and to identify the optimal, multifunctional sorbents to protect against PFAS. A common strategy in environmental remediation research is to use composites comprised of different sorbent materials to enhance the affinity for target chemicals. Studies are ongoing to characterize edible, multifunctional sorbents that can serve as PFAS enterosorbents.

Table 2 summarizes the key findings of this study, highlighting the PFAS accumulation in maternal reproductive systems, their transfer to offspring through milk, systemic and hepatotoxicity in the offspring, as well as the most effective sorbents for ameliorating PFAS toxicity. Short-term oral administration of GRAS sorbents presents a promising preventive intervention to block the exposure and subsequent toxicity of PFAS during susceptible windows. Beyond prevention during exposure, sorbents could also serve as post-exposure treatments because PFAS undergo enterohepatic recirculation and re-enter the gut, where they are accessible to sorbent gastrointestinal sequestration. While this rodent study provides proof-of-concept, further toxicokinetic and long-term safety studies are needed to evaluate the potential of sorbent interventions for human application. Future studies are also warranted to explore sorbent combinations for different scenarios of PFAS exposure. To study PFAS mixture at equal toxicity, future studies are warranted to study PFAS at their individual point of departure in the mixture.

4. Conclusion

This study investigated maternal exposure to PFAS during gestation and lactation and their impact on two generations. PFAS showed reproductive toxicity by accumulating in the ovary and testis of adult SD rats. Maternal milk was shown as an important vehicle for PFAS

to transfer to offspring, and the liver was a major site of PFAS accumulation in offspring. PFAS exposure also induced systemic inflammation, liver damage, and immunosuppression in offspring. These effects have significant implications for long-term health, as early-life exposure to PFAS may make individuals susceptible to chronic diseases. The ability of different sorbents, especially AC and APM, to mitigate these changes indicates their potential as a short-term preventive intervention to reduce the adverse health outcomes associated with PFAS exposure. These results emphasize the urgent need for targeted interventions to mitigate PFAS toxicity, particularly during vulnerable developmental stages. Enterosorbent treatments represent a promising mitigation strategy, with the potential for differential optimization based on chemical profiles to further enhance efficacy.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability

Data will be made available on request.

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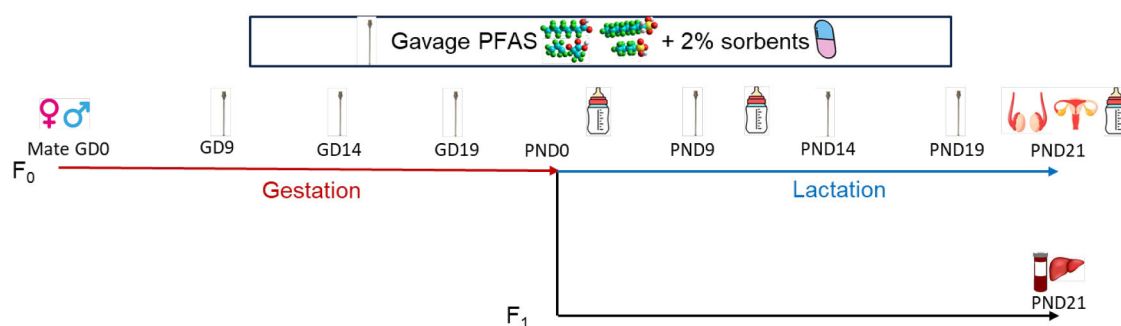


Fig. 1. Schematic diagram of study design. F₀ male (n=4/group) and pregnant female (n=5-7/group) Sprague-Dawley rats were dosed via oral gavage on gestation day (GD) 9, 14, 19, and postnatal day (PND) 9, 14, 19, with ovary and testis collected on PND 21. Milk from F₀ dams was collected on PND 1, 10, and 21. The serum and liver from F₁ offspring were collected on PND 21.

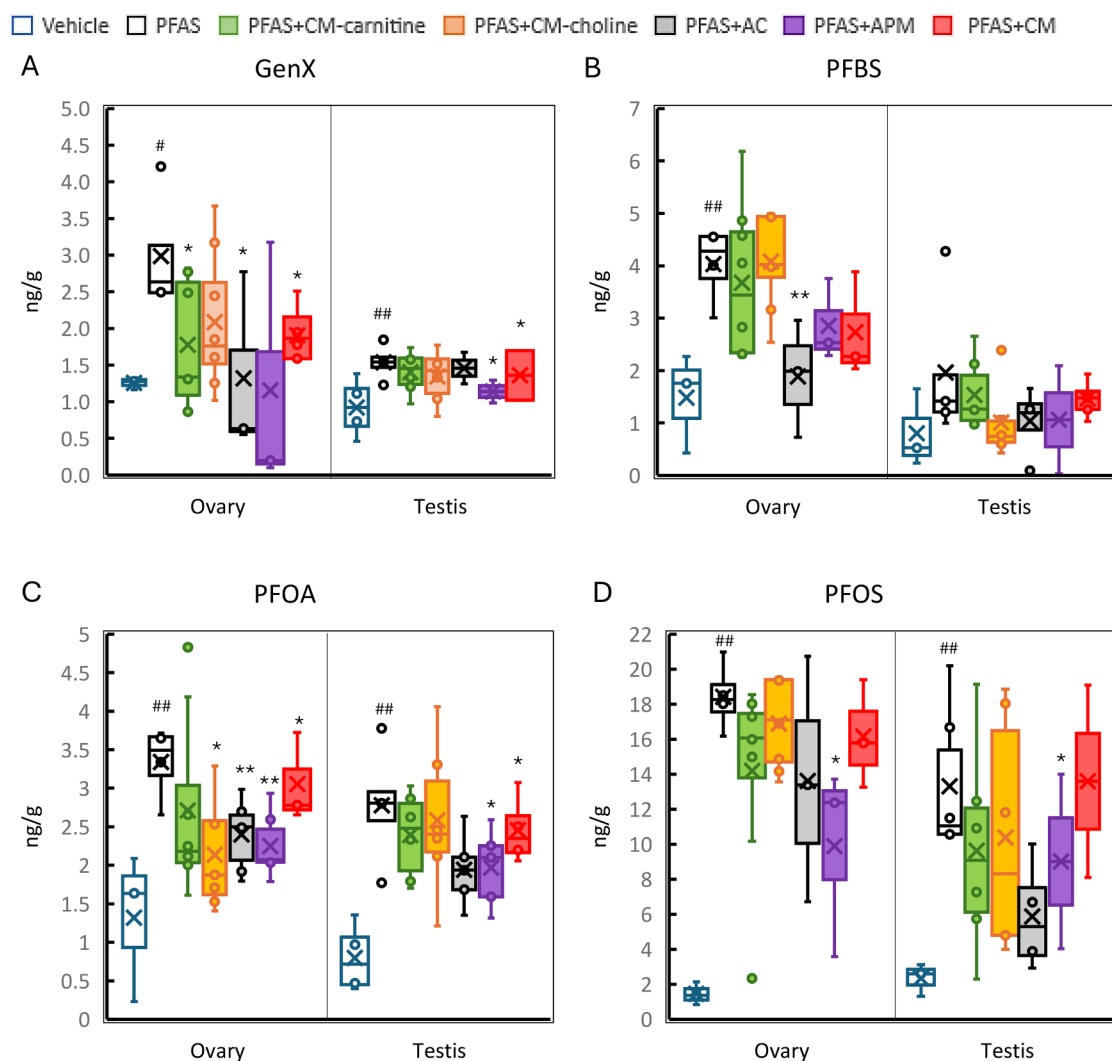


Fig. 2. Detection of GenX (A), PFBS (B), PFOA (C), and PFOS (D) in F₀ ovary and testis on PND 21 ([#]p≤0.05, ^{##}p≤0.01 compared to vehicle control; *p≤0.05, **p≤0.01 compared to PFAS exposure).

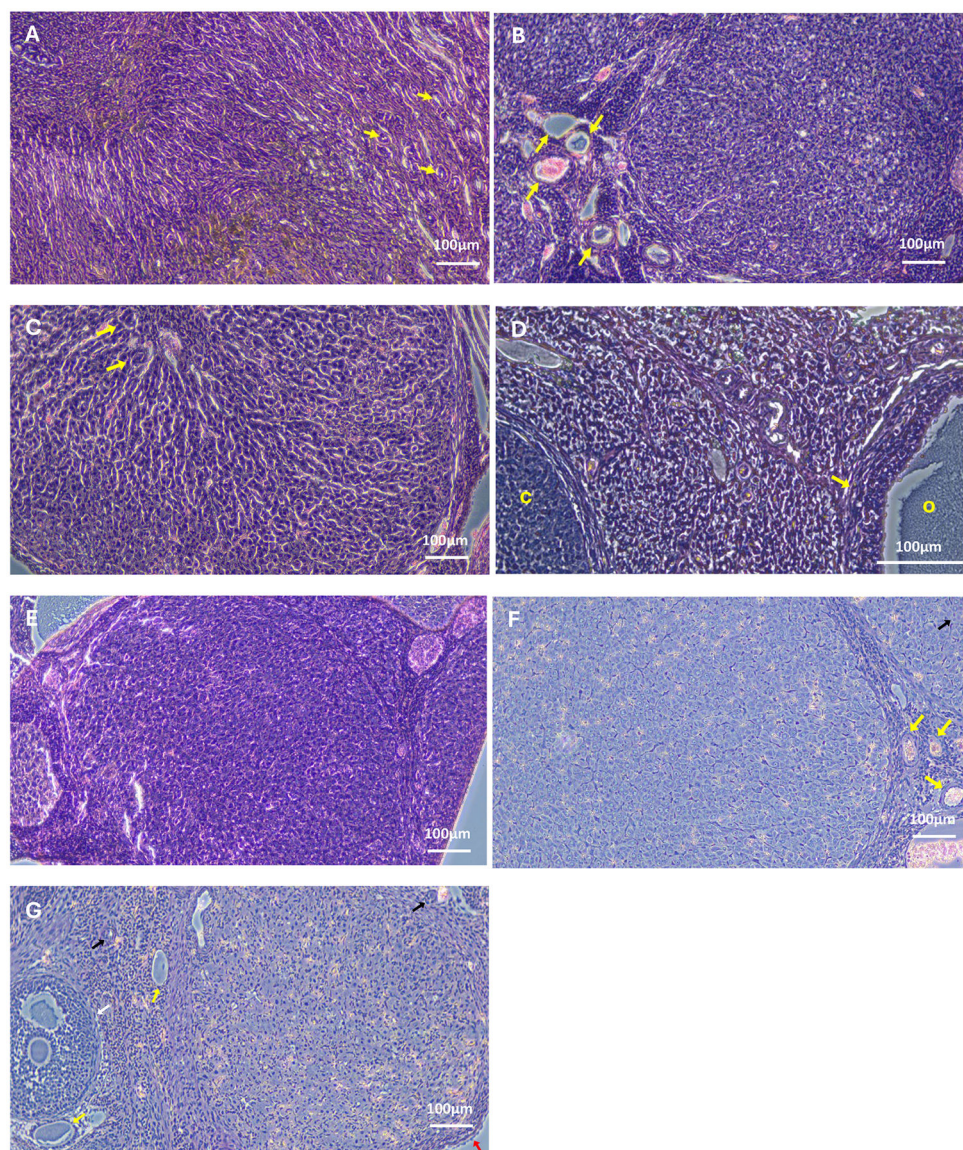
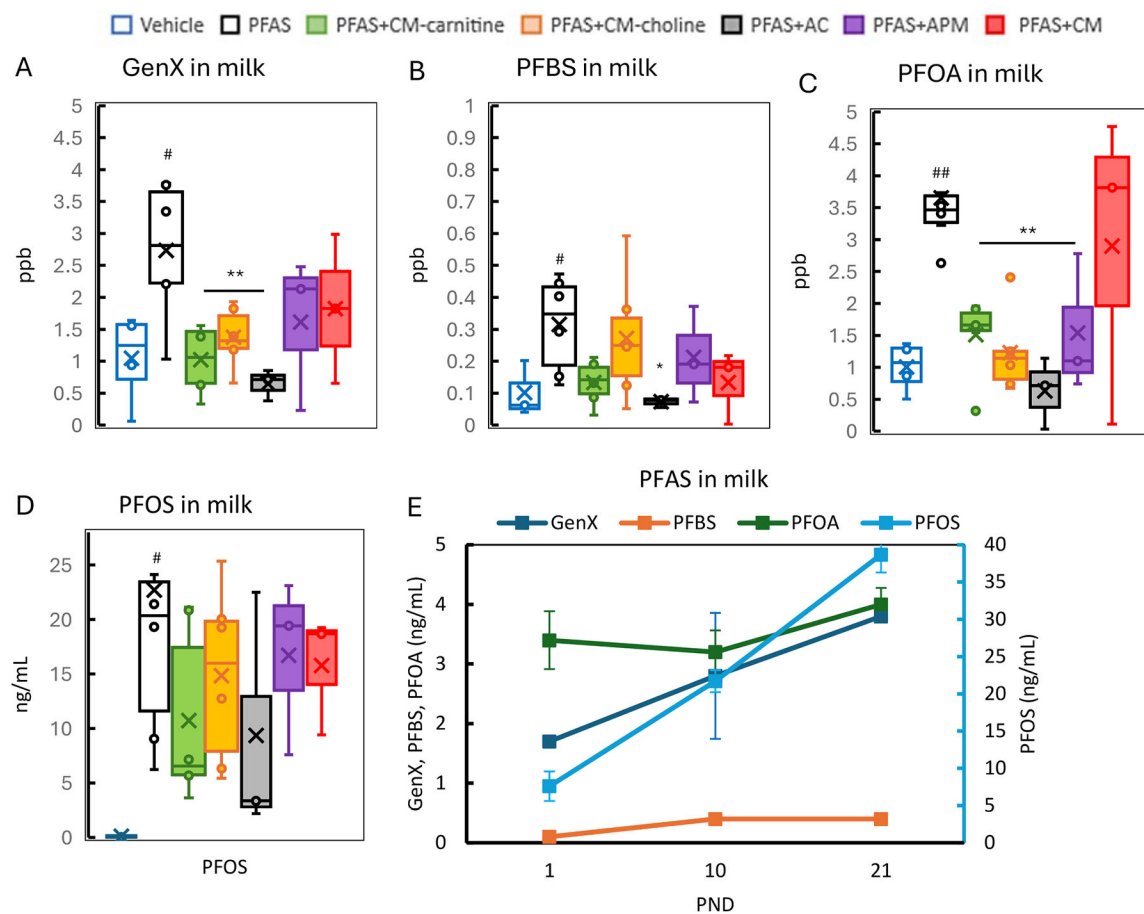


Fig. 3.

H&E-stained ovarian sections from a representative rat in the vehicle control group (A, 20x) showing several observable ovarian follicles (arrows) at various stages of follicular maturation with multiple corpora lutea. The oocytes were intact, with normal histological profiles of the granulosa and theca cell layers. PFAS inhibit follicular growth (B, 20x), showing ovarian follicles in early developmental stages, with a few corpus lutea observed. Many large subcapsular cystic follicles were present, along with dilated and congested blood capillaries (arrows). Treatments with 2% AC (C, 10x), APM (D, 20x), CM (E, 10x), CM-carnitine (F, 20x), and CM-choline (G, 20x) show different degrees of protection.

**Fig. 4.**

Average GenX (A), PFBS (B), PFOA (C), and PFOS (D) in milk with data combined from PND 1, 10, and 21. Detected GenX, PFBS, PFOA, and PFOS (E) at each time point of collection (PND 1, 10, and 21) (# $p \leq 0.05$, ## $p \leq 0.01$ compared to vehicle control; * $p \leq 0.05$, ** $p \leq 0.01$ compared to PFAS exposure).

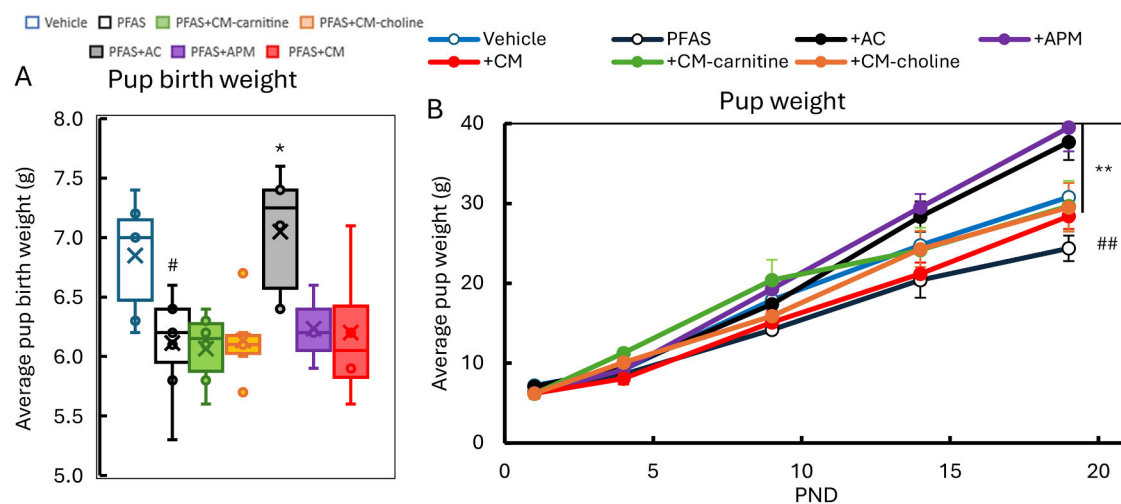


Fig. 5. Average pup weight at birth on PND 1 (A), PND 4, 9, 14, and 19 (B). PFAS significantly reduced pup weight at birth and on PND 19 ($^{\#}p \leq 0.05$, $^{\#\#}p \leq 0.01$) compared to vehicle control; All sorbent treatments significantly improved growth ($^*p \leq 0.05$, $^{**}p \leq 0.01$) compared to PFAS toxicity.

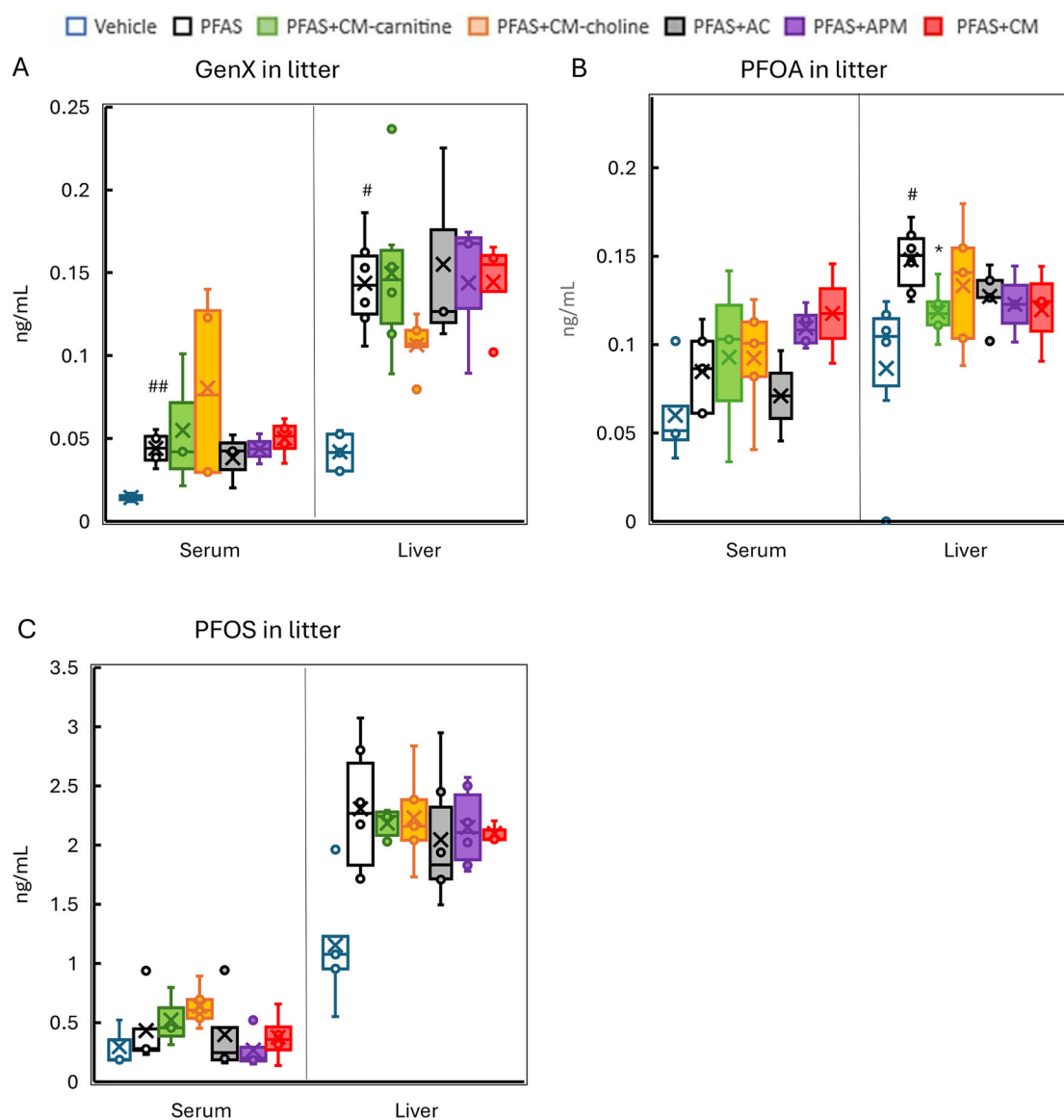
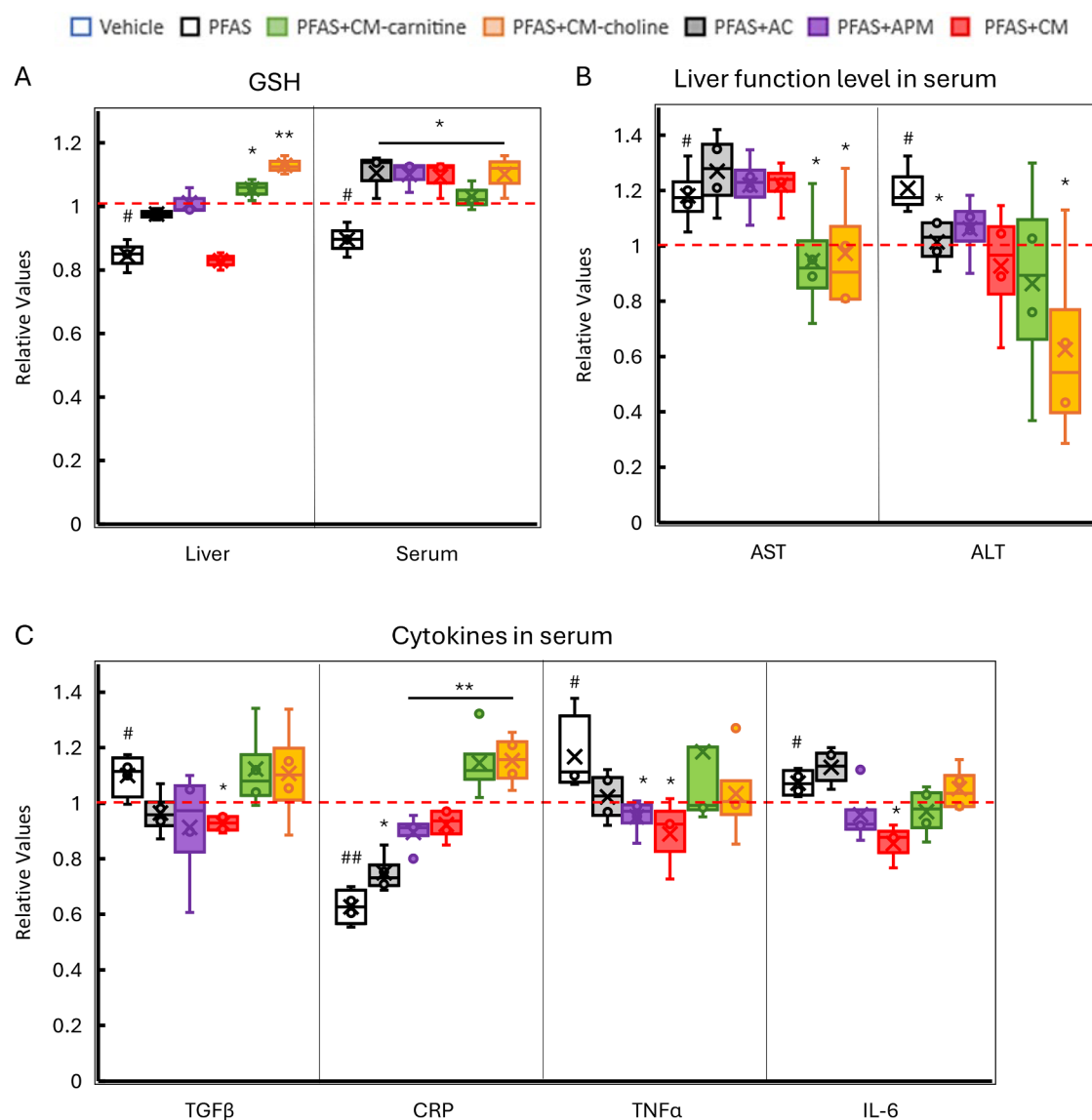


Fig. 6. Detection of GenX (A), PFOA (B), and PFOS (C) in serum and liver of litters on PND 21 (# $p \leq 0.05$, compared to vehicle control; * $p \leq 0.05$ compared to PFAS exposure).

**Fig. 7.**

Expression of GSH in F₁ liver and serum (A), liver function levels of AST and ALT in serum (B), and inflammation-associated parameters of TGF- β , CPR, TNF- α , and IL-6 (C) in serum. All values are relative to the vehicle control, which was adjusted to 1 (red dotted line) (# $p \leq 0.05$, ## $p \leq 0.01$ compared to vehicle control; * $p \leq 0.05$ compared to PFAS exposure).

Table 1Absolute and relative tissue weight of the ovary and testis in F₀ rats on PND 21.

	Absolute weight		Relative to body weight	
	Ovary	Testis	Ovary	Testis
Vehicle	1.38±0.42g	3.65±0.16g	0.49±0.15%	0.92±0.04%
PFAS	0.98±0.19g [#]	4.25±0.46g [#]	0.34±0.07% ^{##}	1.15±0.13% [#]
PFAS+CM-carnitine	0.86±0.14g	4.03±0.28g	0.36±0.06%	1.13±0.08%
PFAS+CM-choline	1.04±0.31g	3.98±0.12g	0.39±0.12%	1.09±0.05%
PFAS+AC	1.13±0.08g	4.35±0.24g	0.40±0.03%	1.15±0.06%
PFAS+APM	1.18±0.14g [*]	4.00±0.12g	0.50±0.06% ^{**}	1.04±0.03%
PFAS+CM	0.90±0.12g	3.95±0.07g	0.35±0.05%	1.08±0.02%

[#]
p≤0.05,^{##}
p≤0.01 compared to vehicle control;^{*}
p≤0.05,^{**}
p≤0.01 compared to PFAS exposure

Table 2

Summary of key findings.

	Endpoint	Impact of PFAS	Best sorbent
F ₁ Females	Relative ovary weight	↓	AC
	Ovary histopathology	Histopathological impact	AC & APM
	PFAS in the ovary	↑↑, PFOS>>others	AC & APM
	PFAS in milk	PND21>10>1	AC
F ₁ Males	Relative testis weight	↑	APM
	PFAS in testis	↑, PFOS>>others	AC & APM
F ₀ pups	Birth weight and growth	↓	AC & APM
	PFAS in serum	↑, PFOS>> others	AC
	PFAS in the liver	↑↑, PFOS>> others	AC
	GSH	↓	CM-choline
	AST & ALT	↑	CM-choline
	TNF-α & IL-6	↑	CM
	TGF-β	↑	CM
	CRP	↓	CM-choline